Aplicação de Lista de Compras com Preços e Alertas

Students:

Alexandre Tavares (Nº 51271)
Email: a51271@alunos.isel.pt
Tel: 968658876
Pedro Gomes (Nº 51423)
Email: a51423@alunos.isel.pt
Tel: 936173635
Martim Monteiro (Nº 51619)
Email: a51619@alunos.isel.pt
Tel: 965420546

Supervisors:

Miguel Gamboa
Email: mcarvalho@cc.isel.ipl.pt

João Pereira
Email:
joao.pereira@primeira-academia.pt
Breno Nico
Email: breno.studart@gmail.com

Pedro Tainha
Email: pedrotainha@gmail.com

Course: Project and Seminary || BSc in Computer & Software Engineering
Data: March 3, 2026

---

1 Introduction

This project proposes the development of a complete software system, from architectural design to publication in app stores and deployment in a public environment. The solution consists of a mobile application for shopping list management, designed for fast real-world usage, allowing users to create and manage lists, consult prices across different stores and locations, and define alerts for price changes.

The application will be supported by an ecosystem of microservices with well-defined REST APIs, developed in .NET, responsible for managing products, prices, and stores, as well as user authentication and authorization through Keycloak. The system will also include push notifications using Firebase Cloud Messaging and a React-based web application.

One of the differentiating elements of the project is the adoption of a *White Label development* model, based on a reusable Design System that will enable the generation of different versions of the application through distinct CI/CD pipelines for build, testing, and deployment.

Development will be supported by Artificial Intelligence tools based on Large Language Models (LLMs), namely GitHub Copilot, while ensuring quality control and technical validation of the produced solutions.

This project integrates mobile, web, backend, and data development, addressing challenges related to distributed architecture, security, data synchronization, and scalability, closely reflecting a real-world professional development scenario.

2 Analysis

2.1 Problems to Solve

The development of a distributed application with a mobile component, backend microservices, and a web backoffice raises several relevant technical and architectural challenges.

1. Data Modeling and Consistency The centralized management of products, prices, and stores requires consistent and normalized data modeling. The existence of geo-referenced prices implies that the same product may have multiple prices depending on location and store, creating challenges in data integrity and efficient updates.

2. Synchronization and Concurrency The application will be used in real-world contexts (e.g., supermarkets), where connectivity may be limited. It is necessary to ensure synchronization between client and server, conflict resolution, and eventual consistency of data.

3. Performance and Scalability Frequent price queries filtered by location may generate significant backend load. The system must be designed to scale horizontally and support multiple simultaneous users.

4. Price Alert System Implementing alerts requires efficient mechanisms for detecting price changes and triggering real-time notifications. It is necessary to ensure reliability, avoid duplicate notifications, and guarantee low latency.

5. Security and Authentication User management, permissions, and integration with external Identity Providers require a secure architecture based on modern authentication and authorization standards.

6. Continuous Integration and White Label The existence of two distinct pipelines (base version and customized version) introduces complexity in automating builds, tests, and deployments. It is necessary to ensure reproducibility and isolation between versions.

7. Quality of Code Generated by LLMs The use of LLM-based tools in development may introduce inconsistent or suboptimal code. It becomes essential to validate, review, and automatically test all produced artifacts.

2.2 Techniques and Tools

To address the identified challenges, the following technical approaches and tools will be adopted:

1. Microservices Architecture with .NET The backend will be developed in .NET, with well-defined REST APIs and clearly separated domains (Products, Prices, Stores, Users). This separation enables independent scalability and improved system maintainability.

2. Relational Database and Geographic Indexing A relational database will be used with proper indexing support for location-based queries. Query optimization techniques and caching strategies may be applied to improve performance.

3. React Native and State Management The mobile application will be developed using React Native, allowing code sharing between Android and iOS. State management will be structured to ensure efficient synchronization between the interface and the backend.

4. Barcode Scanning Barcode scanning functionality will be integrated using the device camera, allowing quick product identification and reducing manual input errors.

5. Geolocation User location will be obtained through native mobile APIs, enabling the display of relevant prices based on proximity.

6. Authentication with Keycloak Authentication will be delegated to Keycloak as an Identity Provider, supporting standard protocols such as OAuth 2.0 and OpenID Connect, ensuring security and interoperability.

7. Push Notifications with Firebase Cloud Messaging Price change alerts will be implemented using Firebase Cloud Messaging, ensuring efficient delivery of push notifications.

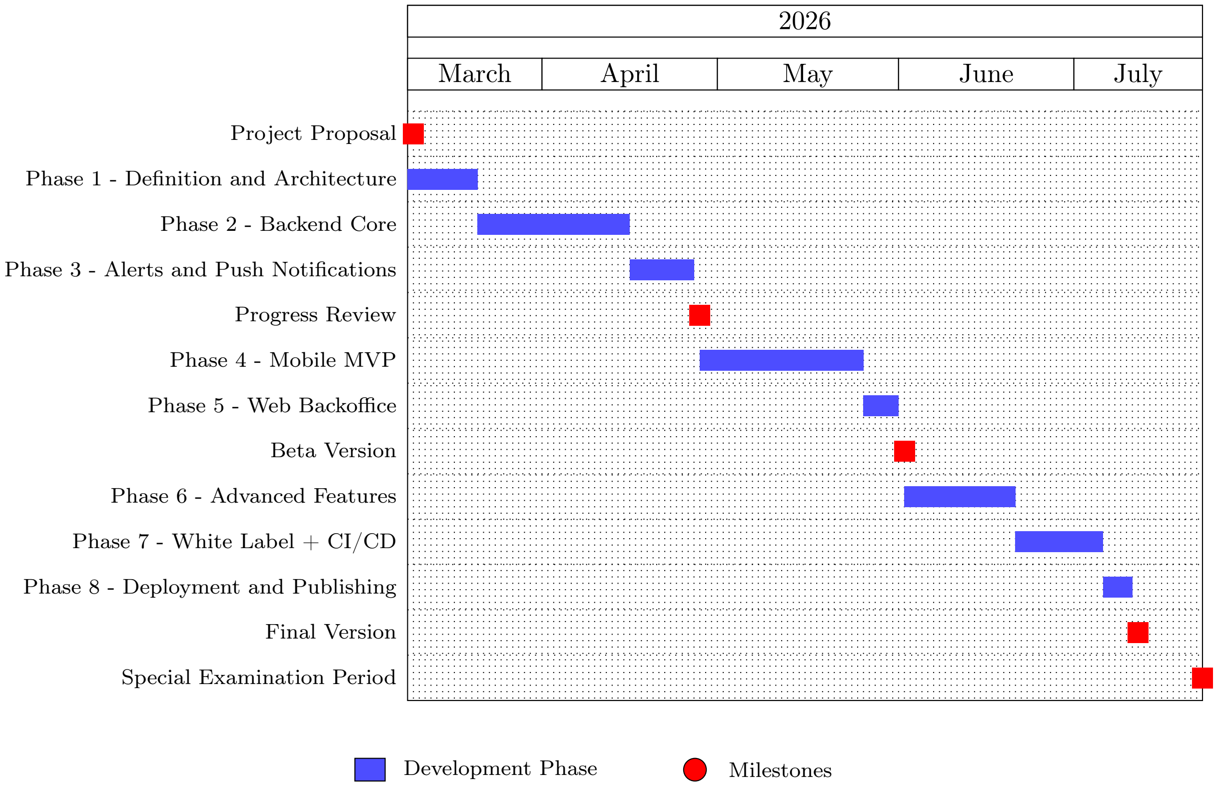
8. CI/CD and Automation Two CI/CD pipelines will be configured: - Pipeline 1: Build, automated testing, and deployment of the base (White Label) version. - Pipeline 2: Design System customization, build, testing, and deployment of the customized version.

These pipelines will automate unit testing, integration testing, and code quality validation.

9. Use of LLMs in Development GitHub Copilot integrated with VS Code will be used to support development. Generated code will be validated through manual review, automated testing, and static analysis to ensure compliance with software engineering best practices.

The combination of these techniques mitigates the identified risks, ensuring that the solution is scalable, secure, and ready for deployment in a real-world environment.

3 Project Plan – Shopping List Application



4 References

- OpenAI. (2024). *ChatGPT (GPT-4/5 series) [Large language model].* Retrieved from https://chat.openai.com/
- Microsoft. (2024). *.NET Documentation.* Retrieved from https://learn.microsoft.com/dotnet/
- React Native Documentation. (2024). *React Native Official Documentation.* Retrieved from https://reactnative.dev/
- Keycloak Documentation. (2024). *Keycloak Server Documentation.* Retrieved from https://www.keycloak.org/documentation
- Fowler, M. (2018). *Microservices: a definition of this new architectural term.* Retrieved from https://martinfowler.com/articles/microservices.html